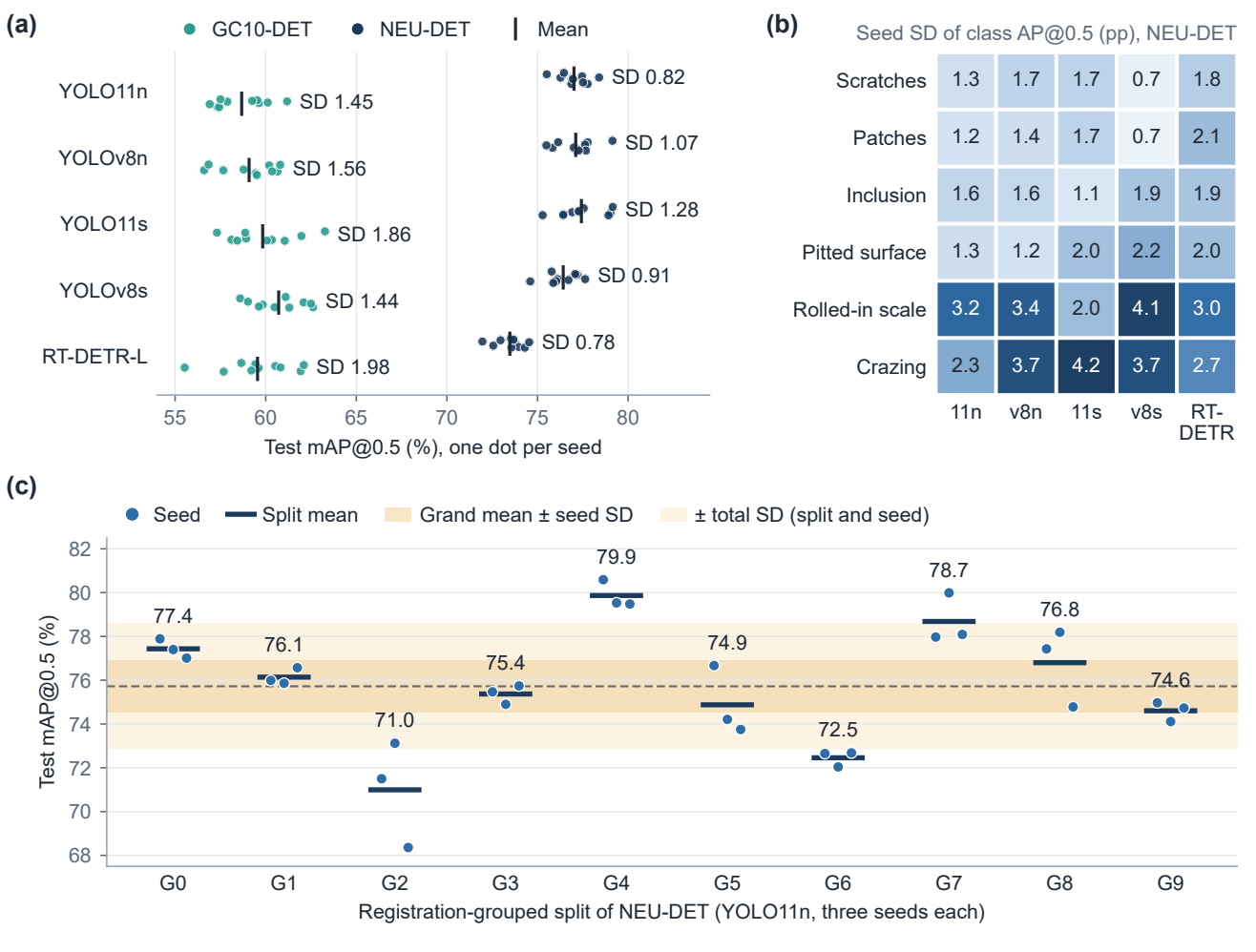

(a)
GC10-DET
NEU-DET
Mean
YOLO11n
YOLOv8n
YOLO11s
YOLOv8s
RT-DETR-L
SD 1.45
SD 1.56
SD 1.86
SD 1.44
SD 1.98
SD 0.82
SD 1.07
SD 1.28
SD 0.91
SD 0.78
55
60
65
70
75
80
Test mAP@0.5 (%), one dot per seed
(b)
Seed SD of class AP@0.5 (pp), NEU-DET
Scratches 1.3 1.7 1.7 0.7 1.8
Patches 1.2 1.4 1.7 0.7 2.1
Inclusion 1.6 1.6 1.1 1.9 1.9
Pitted surface 1.3 1.2 2.0 2.2 2.0
Rolled-in scale 3.2 3.4 2.0 4.1 3.0
Crazing 2.3 3.7 4.2 3.7 2.7
11n
v8n
11s
v8s
RT-DETR
(c)
Seed
Split mean
Grand mean ± seed SD
± total SD (split and seed)
Test mAP@0.5 (%)
82
80
78
76
74
72
70
68
77.4
76.1
71.0
75.4
79.9
74.9
72.5
78.7
76.8
74.6
G0
G1
G2
G3
G4
G5
G6
G7
G8
G9
Registration-grouped split of NEU-DET (YOLO11n, three seeds each)